Main questions

Mystery of training neural networks with (stochastic) gradient descent

- Do they converge to 0 training loss? — non-convex optimisation
- Why do they generalise? — overfitting $\neq$ bad generalisation?
- What is the role of initialisation? — important since non-convex

In this talk:

- 2 layers neural networks
- (Leaky) ReLU activation
- Gradient Flow
- No regularisation

Outline

1. Setting
2. Role of initialisation
3. The early alignment phenomenon
4. Drawback: reaching global minima is hard